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🐙 GitHub Profile

🌐 LinkedIn Profile

EDUCATION

•Bachelor of Computer Engineering

2022-Present

Institut Teknologi Sepuluh Nopember, Surabaya

GPA: 3.64

PERSONAL PROJECTS

•Smart Wheelchair with Eye Movement Control

A smart wheelchair system controlled by eye movement, integrating computer vision and embedded hardware.

- Developed a smart wheelchair system that allows users to control movement using eye gaze, enhancing accessibility for individuals with mobility impairments.
- Integrated real-time eye tracking and gesture recognition using Python and OpenCV, enabling intuitive navigation and control.
- Implemented embedded hardware components (Arduino/ESP32) for motor control and sensor feedback, ensuring reliable operation.
- Collaborated with a team to design, prototype, and test the system, resulting in a functional prototype showcased at the GEMASTIK XVII competition.
- Technology Used: Python, OpenCV, Arduino/ESP32, C/C++, computer vision algorithms.

•Sappy: Multi-Role Livestock Management App (Flutter)

A livestock management app for multiple roles (Farmer, Admin, Doctor) with NFC support.

- Provides role-based access and dashboards for farmers, administrators, and doctors, all in a unified mobile app.
- Supports English and Indonesian languages using built-in localization.
- Implements state management via Provider and enables NFC functionality for secure identification and tracking.
- Includes data visualization, authentication, and environmental configuration using modern Flutter ecosystem packages.
- Technology Used: Flutter, Dart, Provider, NFC, fl_chart, carousel_slider, custom fonts/assets.

•Comic Reading Web Application

A web-based comic reading platform supporting manga, manhwa, and manhua with optimized reading experience.

- Developed a responsive comic reading platform supporting manga, manhwa, and manhua with smooth page navigation and optimized image rendering.
- Implemented core reading features including chapter management, reading history, authentication, and adaptive layouts for multiple devices.
- Technology Used: React (frontend), Laravel (backend/API), PostgreSQL (database), with modular component architecture and optimized asset handling.

•Tree Canopy Detection System

An automated tree canopy detection system using deep learning and drones for efficient forest monitoring.

- Developed an automated tree canopy detection system using deep learning and drones for efficient forest monitoring.
- Curated and annotated a custom dataset of drone-captured forest imagery using Roboflow, implementing manual bounding box labeling for diverse tree species.
- Applied specialized data augmentation techniques (rotation, brightness adjustment, and mosaic) to improve model robustness against varying lighting conditions and aerial perspectives.
- Implemented a YOLOv12 model for high-precision object detection and classification on tree canopies.
- Technology Used: Python, Ultralytics, Roboflow, OpenCV, and Deep Learning.

EXPERIENCE

•Web Developer – MIOT Webpage

Aug 2024- Present

Surabaya

Multimedia and IoT Laboratory (MIOT)

- Designed and developed the official MIOT laboratory webpage to showcase research, publications, and team members.
- Implemented a responsive and modern user interface using React.js and Tailwind CSS, ensuring accessibility across devices.
- Integrated dynamic content management for news, events, and member profiles using JSON-based data structures.
- Collaborated with lab members to gather requirements, iterate on design, and deploy the site using GitHub Pages.
- Optimized site performance and SEO, and provided documentation for future maintenance and updates.

•Research Assistant – Multimedia and IoT Lab (B300)

April - June 2025

Surabaya

Multimedia and IoT Laboratory (MIOT)

- Conducted research on deploying deep learning models on edge devices using NVIDIA Jetson Nano, focusing on computer vision tasks.
- Developed and optimized object detection pipelines using Python and C++ with MobileNet-SSD, YOLOv5n, and YOLOv11s architectures.
- Explored model conversion and acceleration techniques with ONNX and NVIDIA TensorRT for real-time inference.
- Integrated and evaluated models within the NVIDIA DeepStream SDK for efficient video analytics and deployment.
- Benchmarked performance and resource utilization to optimize inference speed and accuracy in constrained environments.

•Lab Teaching Assistant – Basic Programming

Aug 2025 - Nov 2025

Surabaya

Multimedia and IoT Laboratory (MIOT)

- Developed and maintained lab materials, including coding exercises and troubleshooting guides during practicum, while also being responsible for deploying and maintaining the DMOJ online judge website used in coursework.
- Guided students through practical assignments in C/C++, providing real-time support and technical explanations.
- Evaluated student submissions, provided constructive feedback, and contributed to grading rubrics to ensure fair and consistent assessment.
- Organized and led review sessions before exams, addressing common misconceptions and reinforcing key programming concepts.

•Lab Teaching Assistant – Digital Systems

Mar 2025 - May 2025

Surabaya

Robotics and Digital Systems Laboratory (B401)

- Conducted lab sessions for *Rangkaian Digital*, supporting over 100 undergraduate students each semester.
- Developed and maintained lab materials, including hardware schematics and troubleshooting guides for digital logic experiments.
- Guided students through practical assignments in digital circuit design, providing real-time support and technical explanations.
- Collaborated with faculty to improve lab workflows, implement new teaching tools, and enhance the overall learning experience.
- Organized and led review sessions before exams, addressing common misconceptions and reinforcing key concepts in digital systems.

•Lab Teaching Assistant – Telemathics Workshop

Mar 2025 - May 2025

Surabaya

Robotics and Digital Systems Laboratory (B401)

- Prepared and developed practicum modules and instructional materials for *Workshop Telematika*, ensuring structured, hands-on learning for undergraduate students.
- Supervised and guided students during laboratory sessions, particularly in hardware-oriented activities including manual soldering, 3D design for prototyping, hot air rework (blower) soldering, reflow soldering, and 3D printing.
- Provided real-time technical assistance in C/C++ programming and microcontroller-based experiments, helping students troubleshoot both software and hardware issues.
- Assessed student practicum results, delivered constructive feedback, and contributed to grading criteria to ensure fair and consistent evaluation.
- Organized and led review sessions prior to assessments, reinforcing key concepts and practical skills.

POSITIONS OF RESPONSIBILITY

•Team Leader – GEMASTIK XVII (IoT Division)

Jan - Nov 2024

Semarang

Semarang State University (UNNES)

- Led a team representing Institut Teknologi Sepuluh Nopember in the Smart Devices, Embedded Systems, and IoT Division at GEMASTIK XVII.
- Coordinated the team's efforts across software and hardware components, managed timelines, and delegated responsibilities.
- Developed a smart wheelchair system controllable via eye movement, integrating computer vision and embedded hardware.
- Secured the National Finalist award (Juara Harapan) through efficient planning, ideation, and execution of an IoT-based innovation.

•Head of Logistics Department – Multimedia and Game Event (MAGE X)

Jan - November 2024

Surabaya

Multimedia and Game Event (MAGE X)

- Led the logistics team for MAGE X, a major university-level multimedia and game competition event.
- Coordinated procurement, inventory management, and distribution of event materials and equipment.
- Developed and executed logistics plans to ensure timely setup and smooth operation across multiple event venues.
- Collaborated with other departments to address real-time needs and resolve logistical challenges during the event.
- Managed vendor relations, budget tracking, and post-event inventory reconciliation.